Low-degree query lead, step (b): exact rank of the interpolation matrix

Lead: LEAD-EXO-low-degree-query-lower-bound. Protocol protocol.json, LOWDEG-RANK-20261104, was frozen before data (sha256 bbcdab319667..., commit 3edaed22), with the run script, rank routine and analysis pinned by hash. Seeds use the export checker's own derive_seed, and the export passes check-export with 0 findings.

Verdict (frozen rule): HOLDS, with valid controls. With random multipliers b_i, the evaluation matrix of degree-≤D monomials in the coordinates of ([b_1]R, ..., [b_t]R) has **full rank on every candidate log** once the monomial count is about twice the number of candidates. This held in 39 of 39 decisive cells, at p = 251, 509, 1021 and 2039. So the **rank premise** of the desk note's refutation of H1 holds at toy size. The refutation itself is asymptotic. At the tested points the attack spends about 180–490 group operations, more than √N (32–45) and √p/D (11–16), so **no toy-size violation of H1's query bound is shown**. By the validator's rough accounting, the attack beats √N/D only for D ≥ 3 and N roughly 2^28 (D = 5) to 2^48 (D = 3), and never for D = 2. That rests on full rank persisting at t N^(1/D), which was not tested. (Corrected after independent validation. Claim C1's if_supported text in the frozen protocol overstated what a toy-size HOLDS shows.)

This is not an attack-cost claim. Building such a test uses about N^ω local field operations, and the test polynomial has about N/2 or more monomials (see the time-charging note below), so a plain DL table dominates it.

Design

- **Instances.** 37 in total. Two prime-order curves (distinct orders) are taken at each of p = 251, 509, 1021, with D ∈ {2, 3} and t chosen so that C(2t+D, D)/(N−1) is about 0.5, 1 or 2. One instance at p = 2039 has D = 3 and ratio about 2.
- **Arms, with 3 replicates each.**
  - rand: random b_i (the candidate).
  - ap: small b_i = 1..t (reference).
  - shift: X_i = R + [i]P (positive control; rank ≤ 3Dt + 1 by Riemann–Roch).
  - null: uniformly random points of F_p^(2t) (negative control).
- **Decision rule.** HOLDS iff the full-rank fraction of rand cells with ratio ≥ 1.5 is at least 0.9 at every size. The controls must also hold: null full rank and shift within its bound.

Results

size	p	full-rank fraction, rand, ratio ≥ 1.5
L=8	251	1.00
L=9	509	1.00
L=10	1021	1.00
L=11	2039	1.00

- **Controls.** The null is full rank in 100% of its cells, and every shift cell is within 3Dt + 1, far below full rank.
- **Small multipliers (ap).** These are full rank at ratio ≥ 1.5 in 36 of 39 cells. The 3 failures are exactly the smallest-multiplier cells: replicate 0 (b_i = 1..t), D = 3, with Σ b_i² = 0.34–0.41 N. That is consistent with H1′, where the pulled-back degree Δ bounds the rank. 4 of 7 such cells are still full. Every cell with Σ b_i² ≥ 2.1 N is full.

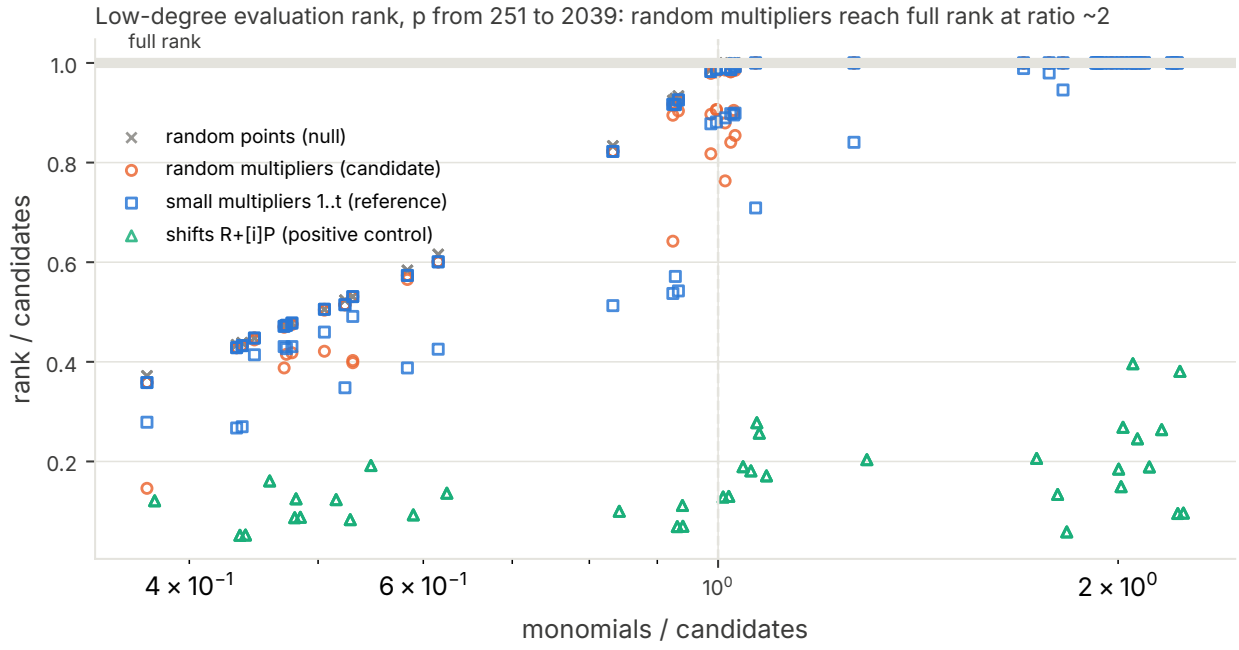

Figure 1: rank against monomials

Post-data observations (not preregistered)

1. **Deficits below ratio 1.5.** Random-multiplier cells fall below the generic expectation min(C, #mon − t·C(2t+D−3, D−3)) at ratios below 1.5, with a mean deficit of 68 at p = 1021 in the 0.75–1.5 band. The round-2 derivation check D2 (docs/reports/leads-round2-20261104/checks/d2.json) explains the p = 211 deficits exactly: every deficient triple carries a short relation Σc_i b_i ≡ 0 mod N with |c| ≤ 2, and 88–90% of fresh triples are exact. Only **two-term** relations (b_i ≡ ±b_j or ±2b_j) produce deficits; three-term relations give none. In this run's middle band, two-term relations account for about 70% of deficient cells (15 of 21 in 0.75–1.5, 10 of 14 below 0.75). The rest are unexplained. In the decisive band, 23 of 39 rand cells carry a two-term relation and all are full rank, so relations do not affect the verdict.
2. **Time-charging note (memo heuristic HD2c).** A test answering "k ∈ S?" for |S| ≈ N/2 needs a polynomial from a span of dimension greater than N/2. Its description, and the per-target evaluation, cost at least about N/2 field operations. The attack is therefore a preprocessing attack with advice N^(1+o(1)), and a DL table beats it. The row's model note should charge description length, not only queries.

Deviations

- **No S3 checkpoint.** There is no upload token. With the owner's approval (session 2026-11-04), the pushed branch is the resume record.
- **Compression.** Rows and export are stored as .xz. Uncompressed hashes are in data.sha256. The export's raw-file entry ld_rows.jsonl is derived from rows.jsonl by make_export_rows.py and reproduces its declared hash (e2af7271…).
- **Provenance.** The export's source and host fields are empty. The run used Python 3.13.18 and NumPy 2.5.3 on the session container, from the frozen-protocol commit. The independent validation (../leads-round2-20261104/VALIDATION-review.md) re-ranked 8 cells exactly and confirmed the verdict and the check-export pass.

Reproduce

```
xz -dc rows.jsonl.xz > rows.jsonl     # or rerun (about 25 min):
python3 lowdeg_rank.py --protocol-sha bbcdab31966734404349e193614ef75b9b6bc2cdc0495aa63a5ccb2cce737756 --out rows.jsonl
python3 analyze_lowdeg.py rows.jsonl verdict.json
python3 figures.py; ./build_pdf.sh
```